

Ruixian Hao

M.S. Researcher | GNSS Remote Sensing and Environmental Monitoring

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Profile

M.S. researcher working on GNSS meteorology, GNSS interferometric reflectometry, and applications of low-cost GNSS equipment. Co-author of **11 peer-reviewed journal articles** (3 first-author and 2 second-author), with experience in algorithm development, GNSS data processing, field deployment, embedded Linux, wireless telemetry, and open research software and data.

Education

- M.S. China University of Mining and Technology - Beijing**, Surveying and Mapping Science and Technology Beijing, China
Sept 2025 - present
- Expected graduation in June 2028; research focuses on GNSS meteorology, atmospheric remote sensing, and GNSS reflectometry.
- B.Eng. China University of Mining and Technology - Beijing**, Surveying Engineering Beijing, China
Sept 2021 - June 2025
- Ranked **1st in the major for four consecutive years** and received recommended admission to the master's program.
 - National Scholarship recipient for two consecutive academic years (2021-2023).

Research Experience

- China University of Mining and Technology - Beijing**, Student Researcher Beijing, China
Sept 2021 - present
- Conducts research in GNSS meteorology, GNSS interferometric reflectometry, and applications of low-cost GNSS equipment.
 - Develops tropospheric-parameter datasets and models from multi-source GNSS, ERA5 reanalysis, radiosonde, and meteorological observations.
 - Designs low-cost, near-real-time GNSS-IR systems that integrate multi-GNSS receivers, edge computing, wireless transmission, field observations, and continuous water-level retrieval.
- National Key Laboratory of Air-Ground Integrated New Navigation System Technology, Beihang University**, Research Intern Beijing, China
July 2024 - Sept 2024
- Investigated ionospheric-scintillation effects on ADS-B civil-aviation trajectories using dual-frequency GNSS carrier-phase combinations.
 - Removed satellite orbit and clock, tropospheric, and relativistic effects to estimate residual-based scintillation indicators.

Selected Research and Engineering Projects

- Low-Cost Near-Real-Time GNSS-IR Water-Level Monitoring** Ningbo and Beijing, China
Feb 2023 - present
- End-to-end environmental sensing system using affordable GNSS hardware.
- Built the complete acquisition-to-retrieval pipeline: raw GNSS capture, 4G telemetry, Raspberry Pi edge control, server-side decoding, sliding-window spectral analysis, and continuous water-level estimation.
 - Deployed synchronized LHCP/RHCP antennas with an Allynstar UM982 receiver at Ningbo Hydrological Station to complement a float gauge obstructed by sediment during low tide.
 - Project report: <https://haoruixian.github.io/blog/2026/ningbo-gnss-ir-field-experiment/>
- ERA5-SH Global Tropospheric Scale-Height Dataset** Open data and software
June 2023 - Mar 2025
- Hourly 1-degree global dataset derived from ERA5 pressure-level profiles for 2013-2022.
- Estimated scale heights for ZTD, ZHD, ZWD, Tm, PWV, and WVD via exponential fitting and validated the products against **591 radiosonde stations**.
 - Published in *Scientific Data* as first author; released code and Zenodo data with **2,916 combined downloads** as of August 2026.
 - Repository: <https://github.com/HaoRuixian/ERA5-SH-dataset-for-troposphere-parameters-code-for-estimate-and-analysis>
- GMR-Water** Open-source MATLAB software
July 2023 - June 2026
- Multi-GNSS, multi-frequency, and multi-observation software for water-level retrieval.
- Supports GPS, Galileo, GLONASS, and BDS with SNR, carrier-phase, and pseudorange strategies; includes correction, quality-control, visualization, GUI, and batch workflows.
 - Published as a second-author software paper in *GPS Solutions*.
 - Repository: <https://github.com/HaoRuixian/GNSS-IR-Kit>

Research Infrastructure

Multi-Receiver and Multi-Sensor GNSS Experimental Infrastructure

Independently designed platforms for fair receiver evaluation and long-term atmospheric observations.

Beijing, China
Mar 2026 - July 2026

- Built a synchronized four-receiver testbed (UM982, ZED-F9P, mosaic-X5, and K827) with RF splitting, Raspberry Pi 5, UPS, a web control dashboard, NTRIP distribution, and local raw-data archiving.
- Deployed a rooftop dual-antenna station with multiple receiver brands, weather and precipitation sensors, solar power, 4G cloud upload, and an RK3588 Linux edge computer.
- Project report: <https://haoruixian.github.io/blog/2026/low-cost-gnss-multi-receiver-test-platform/>

Academic Presentations

CPGNSS Forum 2026

Oral presentation on real-time GNSS-IR sea-level monitoring using low-cost GNSS receivers, covering data acquisition, edge computing, wireless transmission, and continuous retrieval.

Nanjing, China
Apr 2026

- Conference report: <https://haoruixian.github.io/blog/2026/cpgnss-forum-2026/>

Satellite Navigation 2026 Young Scientists Symposium

Poster presentation: Near-Real-Time GNSS-IR Sea-Level Retrieval Using Low-Cost GNSS Receivers; received one of 10 Outstanding Poster Awards.

Beijing, China
Mar 2026

- Conference report: <https://haoruixian.github.io/blog/2026/satellite-navigation-young-scientists-symposium/>

2025 Comprehensive Academic Annual Meeting on Geodesy and Navigation & Seminar on Geodetic Development

Oral presentation delivered on May 17, 2025: Sea Surface Height Estimation Based on Multi-Frequency, Multi-GNSS, and Multi-Method GNSS Multipath Reflectometry.

Qingdao, China
May 2025

Publications

Only first- and second-author publications are listed. Complete publication list: <https://haoruixian.github.io/publications/>

Hao, R., Yang, F., Li, Z., Zhang, Y., Zhou, L., & Wang, L. (2025). ERA5-SH: A global grided scale height dataset for tropospheric parameters based on ERA5 reanalysis. *Scientific Data*, 12(1), 381. DOI: <https://doi.org/10.1038/s41597-025-04714-5>

Hao, R., Yang, F., Chen, W., Li, Z., Wang, L., Yang, R., & Zhang, Y. (2025). The reflectometry method and accuracy analysis of the sea surface height based multi-observation from BDS [J]. *Gnss World of China*, 50(2), 46-53. DOI: <https://doi.org/10.12265/j.gnss.2024208>

Hao, R., Xu, T., Li, Z., Yang, F., Hao, Z., Tan, J., Gao, Y., & Shu, Z. (2024). Spatial-temporal variation of water vapor scale height and its impact factors in different climate zones of China. *Advances in Space Research*, 74(4), 1576-1585. DOI: <https://doi.org/10.1016/j.asr.2024.05.019>

Yang, F., & **Hao, R.** (2026). GMR-water: A water level retrieval software for multi-GNSS, multi-frequency and multi-observation based on GNSS-MR. *GPS Solutions*, 30(3). DOI: <https://doi.org/10.1007/s10291-026-02111-y>

Yang, F., **Hao, R.**, Wang, H., Wang, L., Li, Z., Zheng, J., Wang, Y., & Chen, R. (2025). An optimal method for GNSS water vapor tomography based on parameter adaptation. *GPS Solutions*, 30(1). DOI: <https://doi.org/10.1007/s10291-025-02006-4>

Selected Honors

- **Outstanding Poster Award** (10 recipients), Satellite Navigation 2026 Young Scientists Symposium - 2026
- **First Prize and 1st place**, Future Information Track, Beijing University Innovation Scenario Challenge - 2025
- **National Scholarship**, Ministry of Education of China - two consecutive academic years, 2021-2023
- Second Prize, 13th University GIS Forum Innovation and Entrepreneurship Competition - 2025
- Second Prize, Qingchuang Beijing Challenge Cup - 2025
- Second Prize, National Surveying and Mapping Discipline Innovation and Entrepreneurship Competition - 2023

Technical Skills

Programming: MATLAB, Python, C; scientific visualization and reproducible data workflows

GNSS and Remote Sensing: RINEX/RTCM processing, RTKLIB, GNSS-IR/GNSS-MR, ERA5 reanalysis, radiosonde validation, Kalman filtering

Low-Cost GNSS Equipment: Multi-GNSS receivers from Unicore, SinoGNSS, Septentrio, CNAV, and u-blox; GNSS antennas from Huaxin and Beitian

Systems and Sensors: Embedded Linux (Raspberry Pi series, RK series), NTRIP, 4G telemetry, edge/cloud data acquisition, and deployment of rain gauges and meteorological sensors